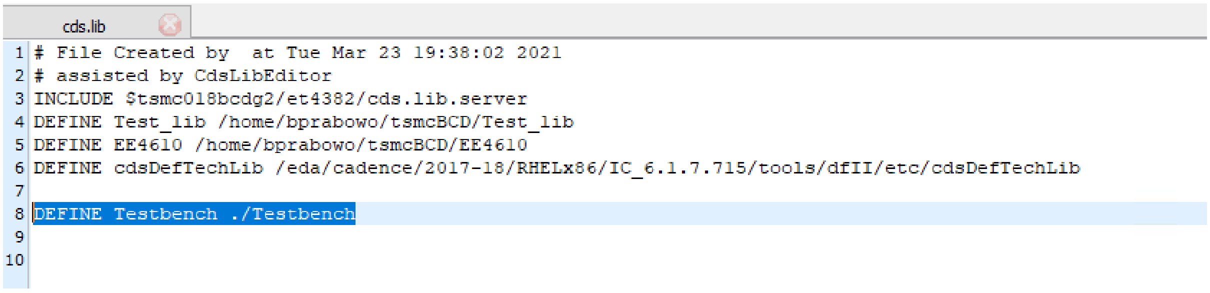
EE4615 Testbench Instructions

Setting up the testbench:

1. Download the testbench to your local PC
2. Extract the testbench zip file to your local PC
3. Upload testbench content to your local director (i.e., the place you start your source sourceme).
4. Check the contents of the extracted folder. The folder should contain all the schematic files and the ocean script.
5. The testbench has to be added to the library of the cadence setup. The easiest way is to add the line:

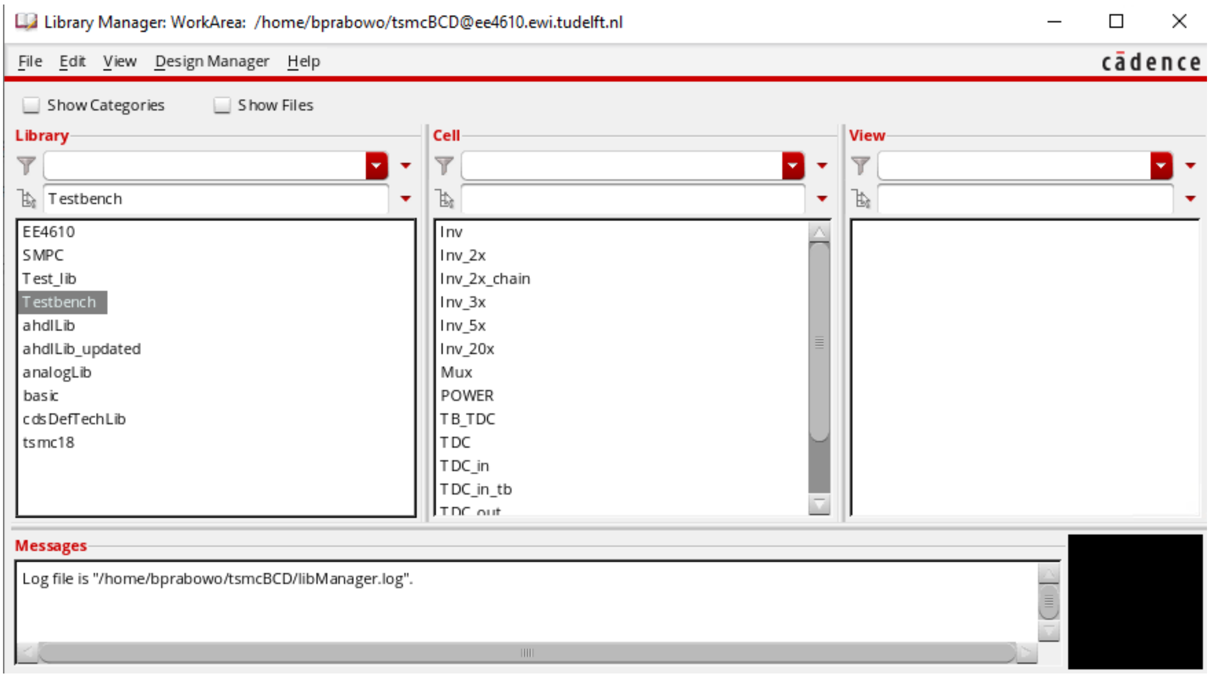
   DEFINE Testbench ./Testbench
   to your cds.lib file



```
# File Created by  at Tue Mar 23 19:38:02 2021
# assisted by CdsLibEditor
INCLUDE $tsmc018bcdg2/et4382/cds.lib.server
DEFINE Test_lib /home/bprabowo/tsmcBCD/Test_lib
DEFINE EE4610 /home/bprabowo/tsmcBCD/EE4610
DEFINE cdsDefTechLib /eda/cadence/2017-18/RHELx86/IC_6.1.7.715/tools/dfII/etc/cdsDefTechLib

DEFINE Testbench ./Testbench
```

6. Save the cds.lib file. You should now see the Testbench added to your libraries.

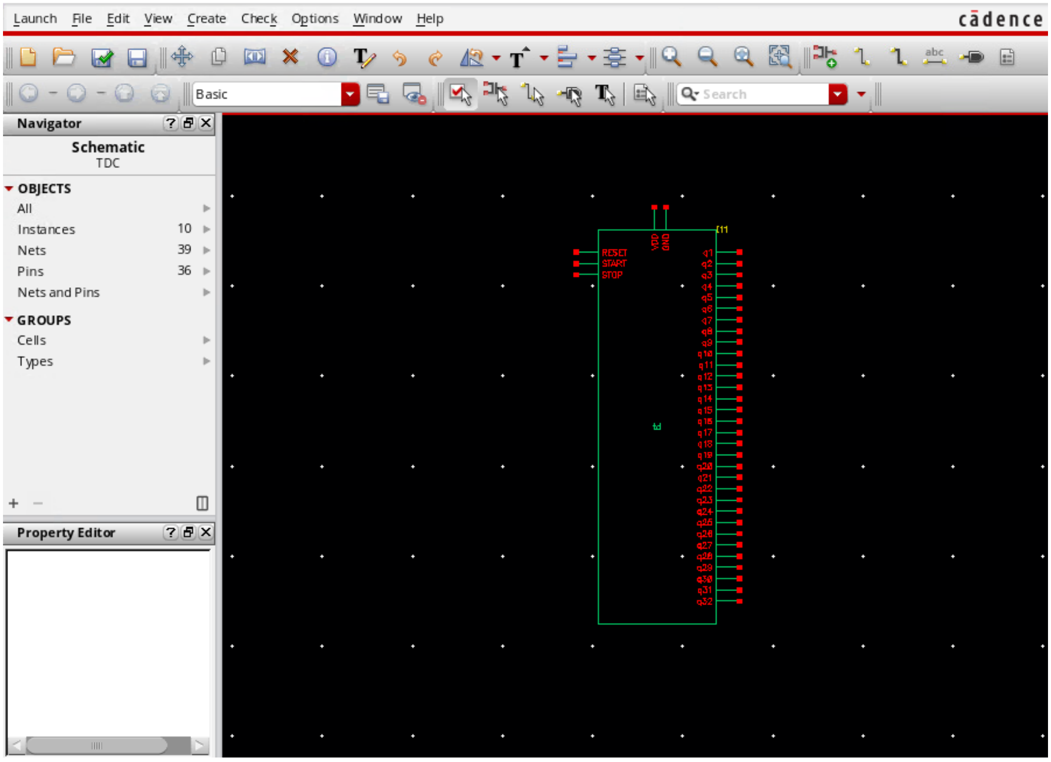


Testbench Modules:

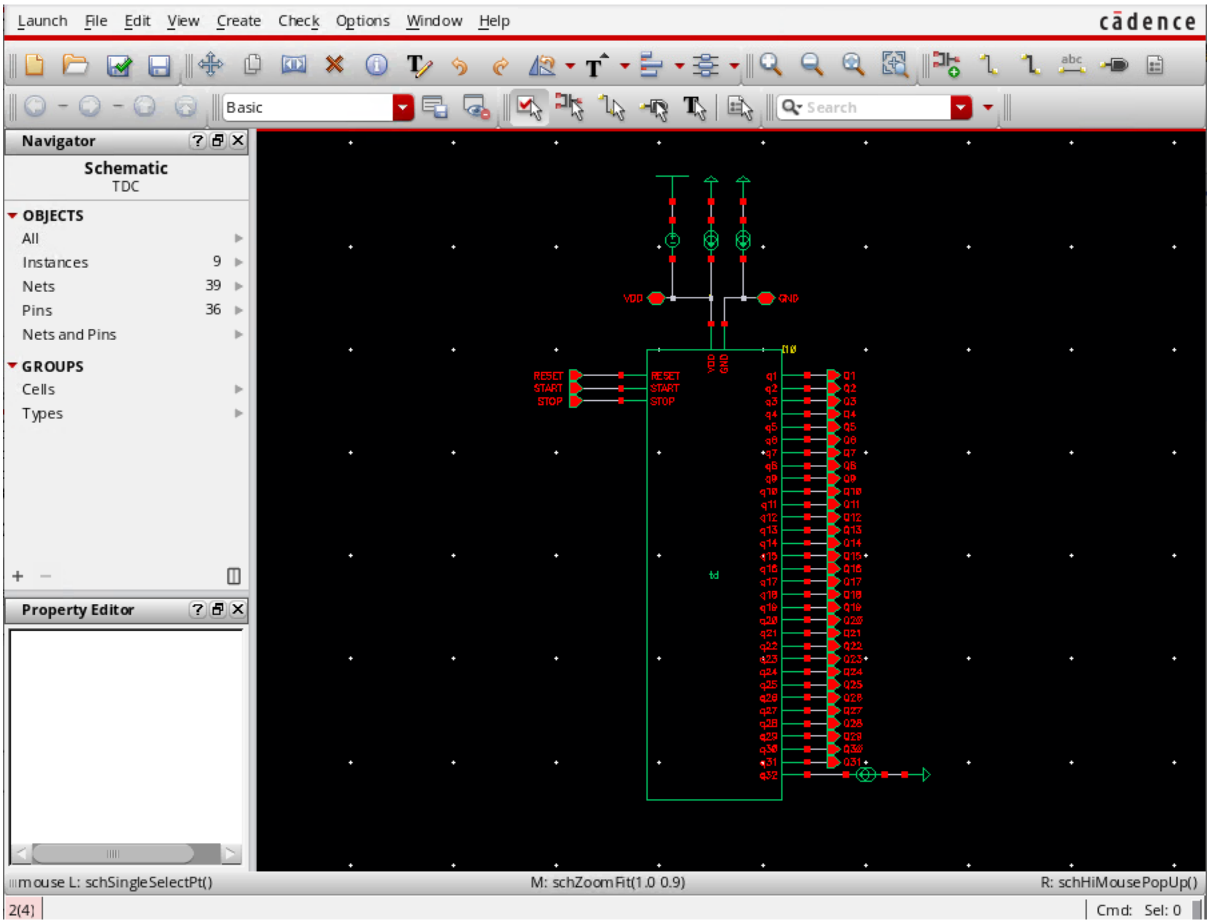
1. **TB_TDC**: This is the top most module of your final design, together with the testbench. It contains the input and output blocks, interfaced with the design unit. It also has a global power supply unit.
2. **TDC_in**: This block generates the input stimuli for the TDC i.e., the RESET, START and STOP signals. Internally these signals are driven using a 3x inverter ('x' being the minimum sized inverter). The RESET signal has a fixed pulse width of 40ps. The signal can be generated synchronously before START and STOP signals.
3. **TDC_out**: This block registers the TDC output
4. **Power**: The power unit supplies power to the entire design. The supply voltage can be adjusted by changing the variable 'supply' in the analog design environment.
5. **TDC**: This forms the top level of your TDC design. The top level of your TDC design has the power signals (VDD & GND) inputs of RESET, START and STOP signals, along with the thermometer/binary output.
6. **5_bit_T2B**: A 5-bit thermometer to binary decoder. **THIS DECODER IS NO LONGER USED TO REDUCE SIMULATION TIME.**
7. **Mux:** Module used to build the 5_bit_T2B decoder.
8. **Txg:** Transmission gates as building blocks of MUX
9. **Inverters:** CMOS based inverters of sizes 'x', '2x', '3x', '5x', '20x' (with 'x' being minimum sized inverters) used as building blocks for other modules.

Interfacing your design to testbench:

1. **Create a symbol of your schematic:** Open your design schematic and create a symbol. Your symbol can have inputs pins for RESET, START, STOP, VDD, GND and 32 output pins which represent your thermometer code outputs as shown in figure below.

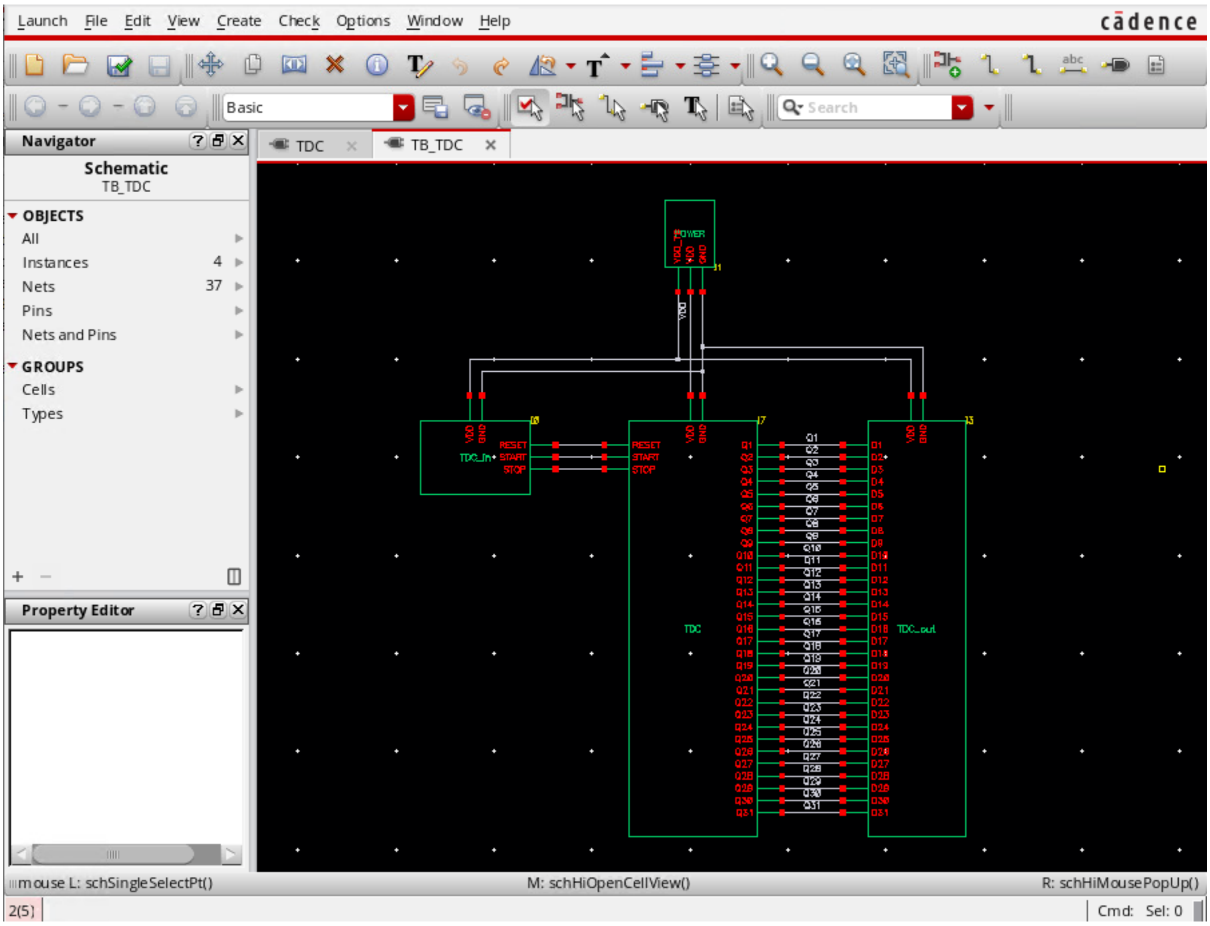


2. **Instantiate your design**: Open the module 'TDC' and instantiate your TDC design. Make sure that you connect the corresponding input pins of RESET, START and STOP correctly in the right order. Don't forget to connect the supply voltages to your design.

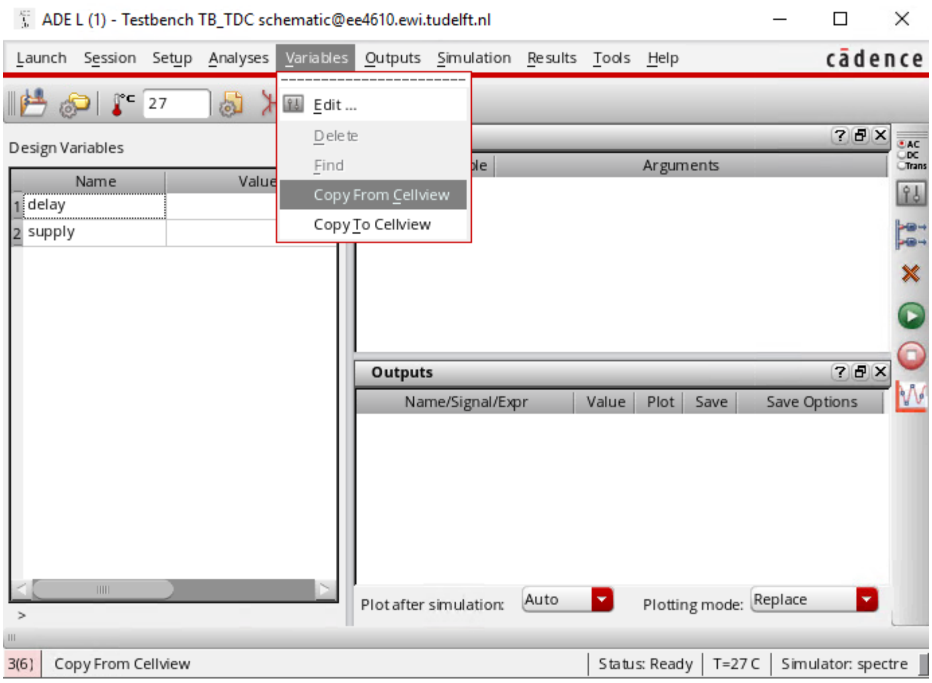


Running the testbench:

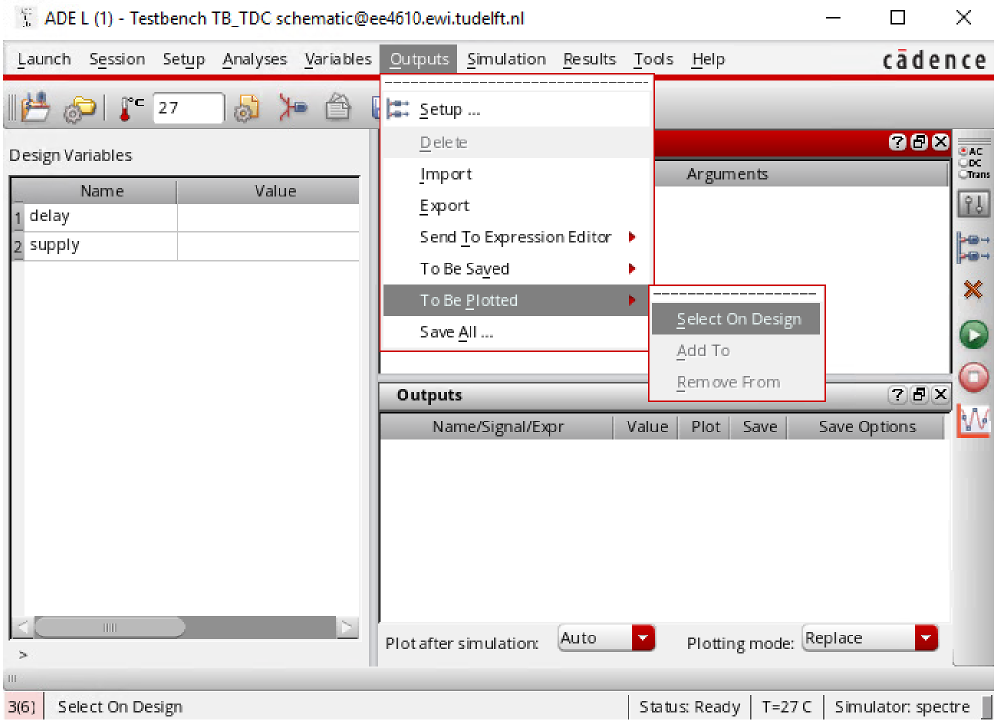
1. Open the 'TB_TDC' schematic file in testbench library.



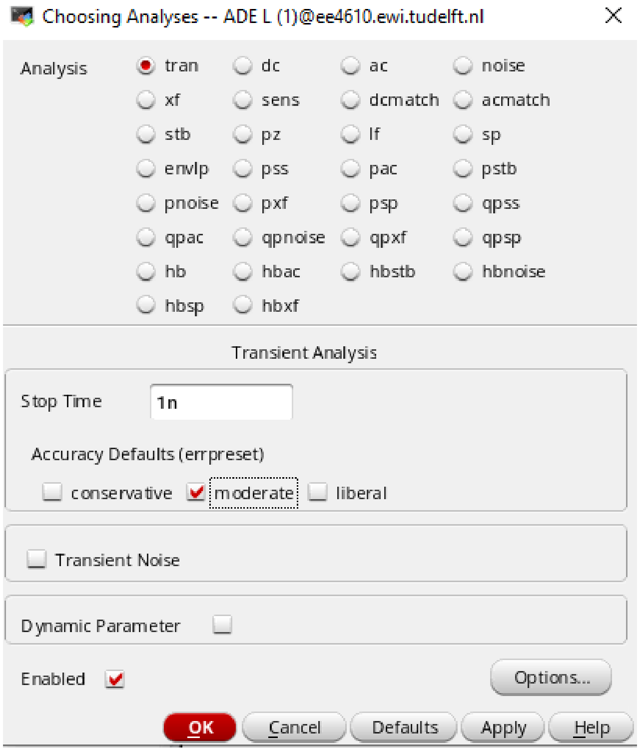
2. Click on 'Launch' => ADE L.
3. Click on 'Variables' => 'Copy from Cellview'



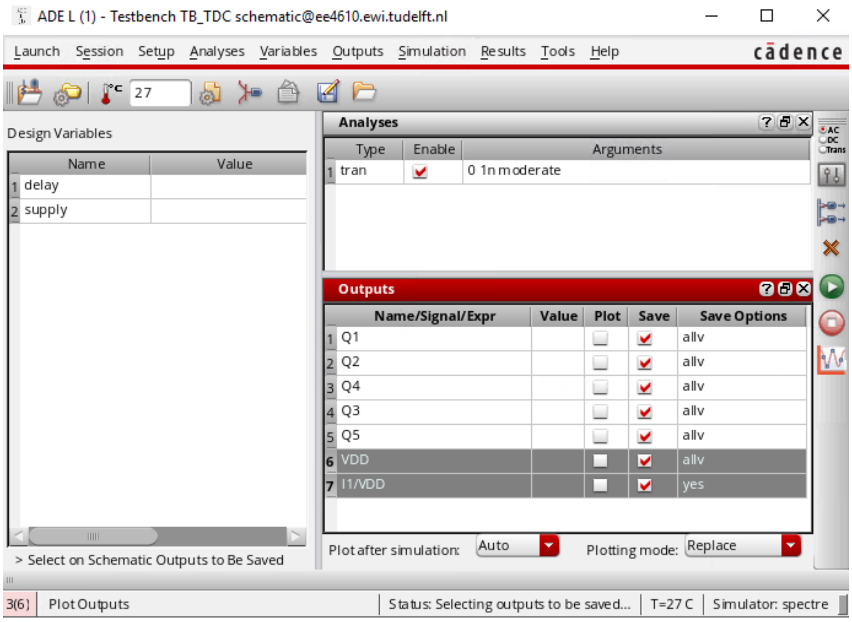
4. Set the variables ‘supply’ and ‘delay’ to your desired value. The ‘supply’ variable refers to the supply voltage of the design, while the ‘delay’ variable refers to the delay between your start and stop signals you wish to set.
5. You can also click on ‘Outputs’ => ‘To be plotted’ => ‘Select from schematic’ and then click the wires or output lines whose waveform you would like to view like that of the inputs RESET, START, STOP and outputs.



6. Now click on analysis, and perform transient analysis with a given simulation time (e.g 1ns)



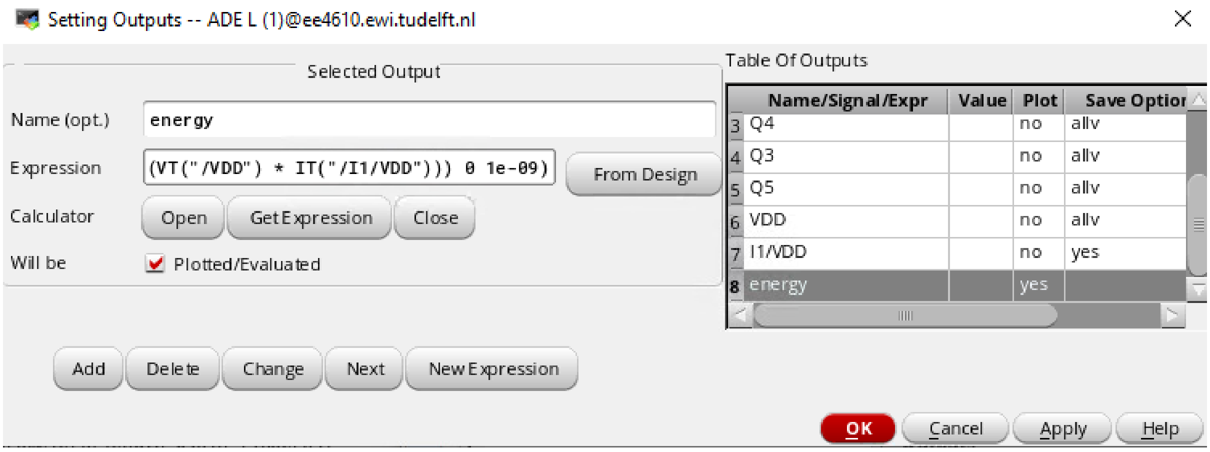
7. In order to estimate the energy, click on 'Outputs' => 'To be Saved' => 'Select on Schematic' and then select the VDD net and the 'red' square on the VDD pin of the POWER block.



8. Make sure that you select both voltage and current. You must see a circle around the red square and the blue wires highlighted. Note that by clicking on the pin (red square) of any signal selects the current, whole clicking on the blue wire selects the voltage of that particular branch. For calculating energy, you need both voltage and current. So, make sure you see a circle around the red square and the blue wire (vdd) being highlighted.
9. Click on ‘Outputs’ => ‘setup’ and enter the name ‘energy’ along with the expression below. Click ‘OK’. You can calculate your energy using the ‘calculator’. Feed in the following formula:

integ((- (VT("/VDD") * IT("/I1/VDD"))) 0 sim_time)

Note: ‘sim_time” is chosen by the user (e.g 1n)



10. You can now perform your transient simulation.

Running Testbench Script:

Two different ocean scripts are provided for binary and thermometer TDC output. The 'ocean command line tool' must be launch using the ocean command in your terminal.

<Work_directory>source sourceme

<Work_directory>ocean

1. Now load the testbench script using the following command to start the parametric sweep of the delay between START and STOP signals.

   <ocean>load("testbench_tdc_therm.ocn")

   Or

   <ocean>load("testbench_tdc_binary.ocn")

2. Use the 'exit' command to exit the command line tool

   <ocean>exit

3. On successfully running the testbench, the two files named 'log.txt' and 'results.csv' are generated in your simulation directory (~/simulation/).

General Remarks:

- Please do not change any schematics files in the TB. In case of any issue please contact the TAs. You are only allowed to change the parametric values such as delay, supply voltages, reset signal pulse widths, timing etc inside the TB.
- Make sure you have a VDD and GND pin in every schematic and symbol of yours to connect them to the top level of your POWER block of the testbench. DO NOT USE ANY OTHER GLOBAL VDD OR GND in your design.